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ABSTRACT

This project develops a low-cost IoT and edge machine-learning node for early detection of spoilage in stored grain. An ESP32 reads three DS18B20 temperature probes (top, middle and bottom of the grain), BME280 humidity sensors, an MQ-135 gas sensor and a grain-moisture probe. It computes a three-level risk score from the vertical temperature gradient, humidity and gas level, signals it with LEDs and a mutable buzzer, logs every reading to a microSD card and publishes live data to a smartphone over MQTT. An Isolation Forest model was trained offline in Python. In simulation (200 samples per scenario) it flagged 3% of normal and 100% of combined spoilage samples, while threshold rules flagged all middle-layer hot-spot samples as high risk. The prototype costs about Rs. 3,357.

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Chapter 1
INTRODUCTION

1.1 BACKGROUND

Food grains such as rice, wheat and maize are harvested once or twice a year but consumed all year, so they spend weeks or months in farm bins, warehouses and household containers, where temperature, humidity and grain moisture decide whether they stay sound or are attacked by moulds and insects.

Moulds and insects respire as they feed, producing heat, moisture and carbon dioxide that favour further growth, so damage accelerates once it starts. Rising temperature in localised hot spots and rising CO2 are among the earliest warning signs [1, 2].

Losses are considerable: an ICAR-CIPHET study estimated annual harvest and post-harvest losses of 45 crops at about Rs. 92,651 crore, and cumulative cereal losses at 4.65% to 5.99% [3].

Traditional monitoring relies on manual smelling and probing, which is labour intensive and intermittent [5]; low-cost Wi-Fi microcontrollers such as the ESP32 now make continuous monitoring practical.

1.2 PROBLEM STATEMENT

Small farmers, village-level storage operators and households usually have no instrument that shows when grain starts to deteriorate. Single-point monitors cannot show where a problem starts, since a hot spot develops in one layer while the rest is normal; in a modelled example, a hot spot 0.9 m from a temperature-cable sensor took about five weeks to be detected [1]. Commercial multi-sensor systems are proprietary and priced for large installations.

The problem is therefore: *how can a low-cost device monitor grain at several depths, detect early spoilage on the device itself, alert the user locally and by smartphone, and keep records when offline?*

1.3 MOTIVATION

The aim is to make early-warning technology affordable at farm-bin scale. The ESP32 [6] and sensors such as the DS18B20 and BME280 [7, 8] are cheap and well documented, and the prototype costs about Rs. 3,357. Machine-learning anomaly detection at the edge also avoids reliance on a single fixed threshold.

1.4 SCOPE OF THE PROJECT

The project delivers a monitoring node and a transparent demonstration container holding a grain sample. It covers three-depth temperature, humidity, gas and moisture sensing, on-device risk scoring, LED and buzzer alerts, offline microSD logging, live MQTT telemetry and an offline machine-learning study on simulated data.

Actuators such as aeration, an on-board display, silo-scale deployment and full-season field trials are out of scope.

1.5 ORGANISATION OF THE REPORT

Chapter 2 states the objectives, Chapter 3 reviews the literature, Chapters 4 and 5 describe the design and implementation, Chapter 6 presents the results and Chapter 7 concludes.

Chapter 2
OBJECTIVES

2.1 AIM

The aim is to design, build and evaluate a low-cost, edge-intelligent IoT node that monitors the micro-climate of stored grain at multiple depths and warns early of spoilage.

2.2 SPECIFIC OBJECTIVES

1. Measure grain temperature at three depths to expose hot spots as a vertical gradient.
2. Measure humidity, gas/VOC level and grain moisture.
3. Compute a risk score on the ESP32 so alerts need no network.
4. Indicate risk with LEDs and a mutable buzzer.
5. Send live readings to a smartphone over MQTT instead of using a display.
6. Log every reading on a microSD card to build a retraining dataset.
7. Train an Isolation Forest model and compare it with threshold rules.
8. Keep hardware cost low.

2.3 DESIGN DECISIONS AND EXCLUSIONS

Three items from the original synopsis were removed: the OLED display (all values go to the smartphone), the 12 V aeration fan (the project focuses on detection and warning, not control) and the fan's current sensor.

Chapter 3
LITERATURE SURVEY AND RESEARCH GAP

3.1 INTRODUCTION

This chapter reviews spoilage factors, IoT and commercial monitoring systems and anomaly detection, and ends with the research gap.

3.2 FACTORS THAT GOVERN SPOILAGE OF STORED GRAIN

Fungal invasion depends mainly on grain moisture, temperature and storage period, and insects respond to the same factors. Their respiration raises CO2, and excess moisture fuels the activity that produces CO2, hot spots and higher temperatures [2].

Temperature cables sense only their immediate surroundings: in a modelled example, a hot spot warming at 3 °C per week took about five weeks to be detected from 0.9 m away [1], so a temperature difference between layers is more informative than one absolute reading.

Gas sensing is also advocated: a recent study reported the potential of early-warning systems built on continuous CO2 monitoring [2]. The MQ-135 used here is not a calibrated CO2 instrument and serves only as a low-cost indicator of gas build-up.

3.3 IOT-BASED GRAIN MONITORING SYSTEMS

Low-cost projects show that grain temperature and humidity can be monitored with microcontrollers [10, 11], including a single-ESP32 system with DHT11, MQ135 and load-cell sensors reporting to Blynk [12] and an ESP32 warehouse system sending MQTT data to a Node-RED dashboard [13].

3.4 COMMERCIAL MONITORING SYSTEMS

Commercial systems such as TeleSense [5] and OPIsystems [9] use wireless temperature, moisture and CO2 sensors with cloud reporting, and TeleSense applies machine learning to detect mould and insect problems. They target large bins and silos, so they are costly and cloud-dependent.

3.5 ANOMALY DETECTION FOR SENSOR DATA

Labelled spoilage data are scarce, so unsupervised anomaly detection suits the problem. Isolation Forest isolates observations by random partitioning, and anomalies need fewer partitions and score lower [14]; it is fast, has few parameters and is available in scikit-learn [15].

3.6 SUMMARY OF THE REVIEWED WORK

Table 3.1: Summary of reviewed systems

System	Sensing and platform	Limitation relative to this project
Rice storage monitor [10]	Temperature, humidity, water content; IoT network	Single measurement point; no on-device anomaly detection reported
Telkom DHT22 system [11]	DHT22, ESP32, LCD, IoT platform	Temperature and humidity only; requires local display
Smart grain monitor [12]	DHT11, MQ135, load cell, ESP32, Blynk	One temperature point; threshold alerts; cloud app needed
Grain warehouse system [13]	ESP32, MQTT, Node-RED dashboard	Warehouse level monitoring; no multi-depth gradient analysis
TeleSense [5]	Wireless temperature, moisture and CO2 sensors; cellular; ML in app	Commercial, proprietary, designed for large bins
OPI system [9]	Wireless cables and CO2 sensor; hourly reports	Commercial, proprietary, designed for large bins

3.7 RESEARCH GAP

Reviewed low-cost systems measure at one point with fixed thresholds, while commercial systems that sense inside the grain and use machine learning are proprietary and priced for large operators. Little published work combines in one open, low-cost node three-depth sensing, humidity, gas and moisture sensing, on-device risk scoring that works offline, a microSD log for retraining and phone-based visualisation. This project addresses that gap.

Chapter 4

PROJECT DESCRIPTION AND DESIGN

4.1 SYSTEM OVERVIEW

The edge IoT node has three parts. The sensing layer has three DS18B20 temperature probes at different depths, two BME280 modules, an MQ-135 gas sensor and a grain-moisture probe. The ESP32 DevKit V1 reads them every five seconds, computes the risk score, drives the alert unit, logs to the microSD card and publishes over MQTT to a smartphone (Fig. 4.1).

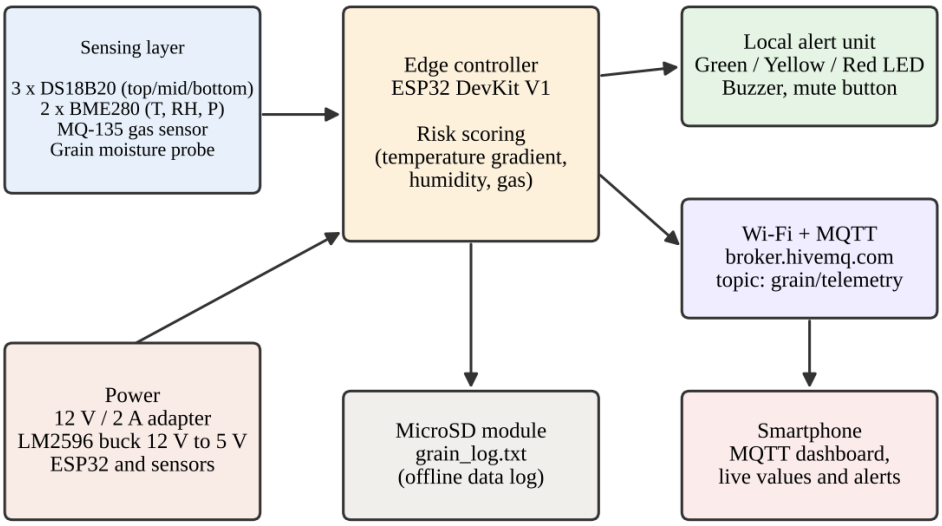


Fig. 4.1: System architecture of the grain storage monitoring node

4.2 HARDWARE DESIGN

4.2.1 Component list

Table 4.1 lists the components with approximate prices including GST. Items 1 to 14 are electronics and items 15 to 18 are mechanical parts and grain; the relay is a spare for future actuator expansion.

Table 4.1: Bill of materials

Sl. No.	Component	Specification	Qty	Unit cost incl. GST (Rs.)	Total (Rs.)
1	ESP32 DevKit V1	WROOM-32, dual-core, Wi-Fi + BLE	1	413	413
2	DS18B20 waterproof probe	Digital, one per layer	3	83	249
3	BME280 module	Temperature + humidity + pressure, I2C	2	295	590
4	MQ-135 gas sensor module	VOC/NH3/benzene air-quality sensor	1	106	106
5	Grain moisture probe	Capacitive/resistive moisture probe	1	142	142
6	1-channel relay module	5 V logic, opto-isolated (spare)	1	83	83
7	12 V DC power adapter	2 A SMPS adapter	1	295	295
8	Buck converter 12 V to 5 V	LM2596 based, adjustable	1	71	71
9	LEDs (green, yellow, red)	5 mm, standard	3	2.4	7.2
10	Active buzzer	5 V, through-hole	1	18	18
11	Push button	6 mm tactile (alarm mute)	2	6	12
12	MicroSD module + 8 GB card	SPI logger module and card	1	236	236
13	Breadboard + jumper wires	Full breadboard, M-M / F-M / F-F set	1	295	295
14	Perfboard/PCB	General purpose, about 7 x 9 cm	1	59	59
15	Plastic project enclosure	ABS box, about 150 x 100 x 50 mm	1	413	413
16	Transparent grain container	Bucket/box for demonstration model	1	177	177
17	Grain sample	Rice, wheat or maize, small quantity	1	50	50
18	Perforated PVC pipes/tubes	Probe sleeves, 3 lengths	3	47	141
	Total (approximate)				**3,357**

The electronics cost about Rs. 2,576 and the remaining parts Rs. 781.

4.2.2 Role of each component

The ESP32 reads the sensors and runs the risk logic. The three DS18B20 probes share one 1-Wire bus; one BME280 monitors the headspace and the second (address 0x77) is a spare ambient reference. The MQ-135 indicates gas build-up from mould or insects.

4.2.3 Pin assignment

Table 4.2 gives the GPIO assignment used in the firmware.

Table 4.2: ESP32 pin assignment

Device	ESP32 pin	Remark
DS18B20 probes (data)	GPIO 4	1-Wire bus, three probes
BME280 (SDA, SCL)	GPIO 21, GPIO 22	I2C, address 0x76
MQ-135 (AOUT)	GPIO 34	Analogue input
Grain moisture probe (AOUT)	GPIO 35	Analogue input
MicroSD module (CS)	GPIO 5	SPI default pins
Green LED	GPIO 12	Low risk
Yellow LED	GPIO 13	Moderate risk
Red LED	GPIO 14	High risk
Buzzer	GPIO 15	High risk alarm
Push button	GPIO 16	Pull-up; mutes buzzer

4.2.4 Physical assembly

Three perforated PVC tubes stand vertically in the transparent container, each holding one DS18B20 probe at the top, middle or bottom. The other sensors are mounted at the upper inner surface, and the electronics sit on a perfboard inside the ABS enclosure (Fig. 4.2).

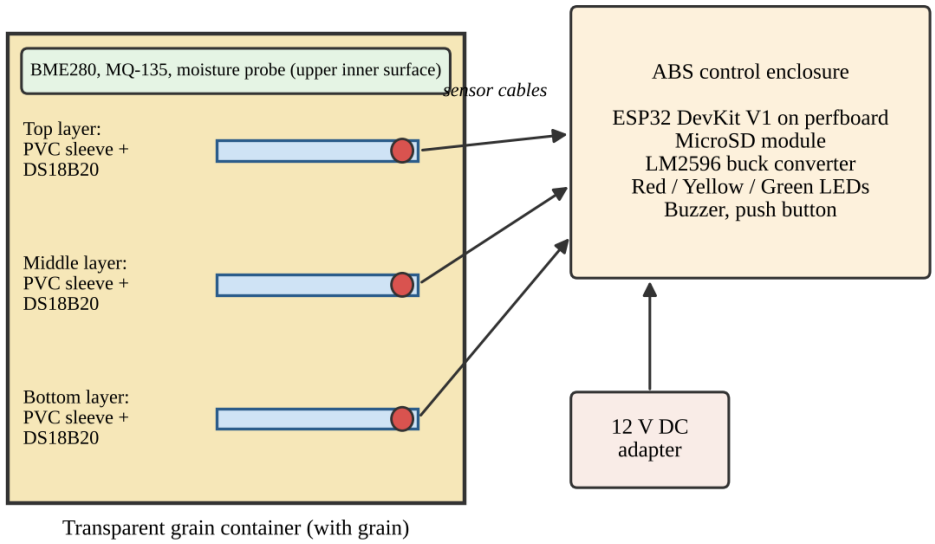


Fig. 4.2: Physical layout of the grain container, sensors and control enclosure

4.3 SOFTWARE DESIGN

4.3.1 Firmware flow

The firmware loops every five seconds: it reconnects to the broker if needed, reads all sensors, calculates the risk score, updates the LEDs and buzzer, appends a line to the SD card and publishes a JSON message (Fig. 4.3).

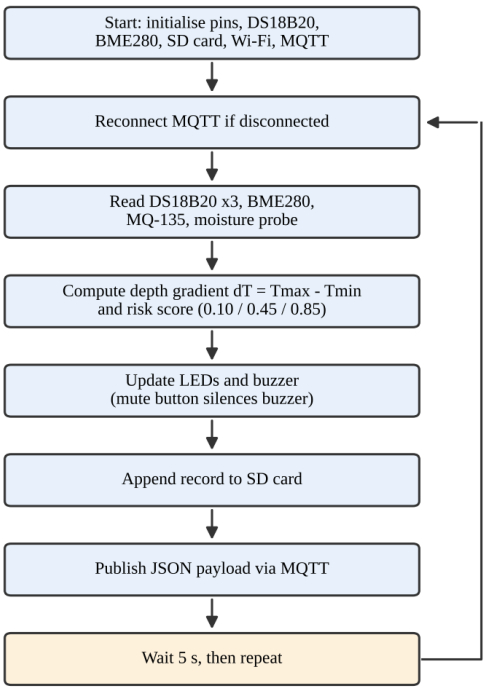


Fig. 4.3: Flowchart of the ESP32 firmware

4.3.2 Risk scoring

The depth gradient is ΔT = T_max − T_min over the three DS18B20 readings; a large ΔT means one layer is warmer than the rest, the signature of a hot spot. Table 4.3 shows how ΔT, humidity and gas set the risk level.

Table 4.3: Risk levels and alert outputs

Risk level	Condition	Score	LED	Buzzer
Low	None of the conditions below	0.10	Green	Off
Moderate	ΔT > 3 °C or humidity > 60 %	0.45	Yellow	Off
High	ΔT > 5 °C or humidity > 70 % or gas ADC value > 2000	0.85	Red	On (can be muted)

A score of 0.7 or more lights the red LED and sounds the buzzer, 0.4 or more lights yellow, and lower scores light green. A button press silences the buzzer until the risk falls below 0.4.

4.3.3 Data formats

Each cycle the node publishes a JSON message (tempTop, tempMid, tempBottom, humidity, gas, risk) to the topic grain/telemetry on broker.hivemq.com, and a status message to grain/status on connection. MQTT is lightweight and suited to constrained devices [16]. The same reading is also written to the SD card as one comma-separated line.

4.4 MACHINE-LEARNING COMPONENT

In parallel, an Isolation Forest is trained offline on a PC using six features: the three layer temperatures, humidity, gas level and grain moisture. Readings that deviate from normal receive negative decision scores (Chapters 5 and 6).

Chapter 5
METHODOLOGY / TECHNICAL IMPLEMENTATION

5.1 DEVELOPMENT APPROACH

The work proceeded from component selection and design simplification through firmware development, model training and integration to evaluation on simulated scenarios.

5.2 HARDWARE IMPLEMENTATION

5.2.1 Wiring and jumper wires

The prototype is first wired on a breadboard using 36 jumper wires (22 female-to-male and 14 male-to-male).

5.2.2 Why a perfboard is used

Because breadboard contacts loosen when moved, the circuit is soldered on a perfboard of about 7 x 9 cm with common 5 V, 3.3 V and ground buses, and fitted in the enclosure.

5.3 FIRMWARE IMPLEMENTATION

5.3.1 Development environment and libraries

The firmware is written in C++ in the Arduino IDE with the board set to ESP32 Dev Module. Table 5.1 lists the libraries; WiFi.h, SPI.h and SD.h come with the ESP32 board package.

Table 5.1: Arduino libraries used

Library	Author	Purpose
OneWire	Paul Stoffregen	1-Wire protocol for the DS18B20 probes
DallasTemperature	Miles Burton	Reading several DS18B20 sensors on one bus
Adafruit BME280 Library	Adafruit	Temperature, humidity and pressure from BME280
Adafruit Unified Sensor	Adafruit	Dependency of the BME280 library
PubSubClient	Nick O'Leary	MQTT client for telemetry

5.3.2 Main program structure

setup() configures the pins, sensors and SD card, connects to Wi-Fi and sets the MQTT broker. loop() follows Fig. 4.3, taking the gradient from the max and min of the three probes and the risk score from the model header file.

5.3.3 Alarm mute button

The push button on GPIO 16 uses the internal pull-up. Pressing it switches the buzzer off while the red LED stays on; the mute clears when the risk falls below 0.4.

5.4 MACHINE-LEARNING MODEL

5.4.1 Training data

No field dataset was available, so the model was trained on 1,000 synthetic samples of six features drawn from the normal distributions in Table 5.2, representing healthy storage. The gas values are in arbitrary units, not MQ-135 ADC counts.

Table 5.2: Parameters of the synthetic baseline (normal) data

Feature	Mean	Standard deviation
Top-layer temperature (°C)	25.0	1.5
Middle-layer temperature (°C)	24.5	1.0
Bottom-layer temperature (°C)	23.5	1.0
Relative humidity (%)	55.0	3.0
Gas / VOC level (arbitrary units)	120	15
Grain moisture (%)	13.5	0.8

5.4.2 Model training

An Isolation Forest with 50 trees and an assumed contamination of 0.05 was fitted with scikit-learn [15] and a fixed random seed, so about 5% of training samples are labelled anomalous. A normal example reading scored +0.157, while a spoilage reading with a mid-layer hot spot (34.5 °C) and VOC spike scored −0.131.

5.4.3 Deploying the model on the ESP32

A microcontroller cannot run a Python pickle file, so the model had to be converted to C/C++:

1. **m2cgen:** failed with NotImplementedError because it does not support IsolationForest.
2. **emlearn:** failed on the Python 3.14 installation because the distutils module was missing.
3. **Header-generation script:** a standalone joblib script generates ml_anomaly_model.h with get_anomaly_score(), applying the rules of Table 4.3 and returning 0.85, 0.45 or 0.10.

Consequently the ESP32 currently runs the threshold scoring in the generated header, not the tree ensemble; exporting the Isolation Forest to the device is future work.

5.5 DATASET GROWTH AND RETRAINING

Every five seconds the ESP32 appends a line to grain_log.txt on the microSD card, building a dataset for the specific container, crop and conditions. The proposed loop is to collect the log, retrain on a PC, regenerate the header and re-flash the ESP32; automating it through MQTT, a database and OTA updates is a design proposal, not implemented.

5.6 EVALUATION METHODOLOGY

The model and threshold rules were compared on simulated scenarios of 200 samples each, generated with a different seed from the training data: a normal set (Table 5.2) to measure false alarms, and four spoilage scenarios (Table 5.3).

Table 5.3: Definition of the simulated scenarios (200 samples each)

Scenario	Change from baseline (mean, standard deviation)
Normal (held-out)	Baseline distribution of Table 5.2
Mid-layer hotspot	Middle temperature 34.0 °C, SD 2.0
Humidity trapping	Humidity 72 %, SD 3; grain moisture 16 %, SD 1
Gas build-up	Gas level 320, SD 40
Combined spoilage	Middle temperature 33 °C (SD 2), humidity 68 % (SD 3), gas 310 (SD 40), moisture 18 % (SD 1)

The Isolation Forest flags a sample when its decision score is negative. The rule detector flags high risk when $\Delta T > 5$ °C, humidity > 70% or gas > 300 (synthetic scale), and moderate or high when $\Delta T > 3$ °C or humidity > 60%.

Chapter 6

RESULTS AND DISCUSSION

6.1 NATURE OF THE RESULTS

These results come from the simulation of Section 5.6, not field measurements; performance on real grain needs a storage trial with the prototype.

6.2 DETECTION PERFORMANCE

Table 6.1 and Fig. 6.1 give the percentage of samples flagged by each detector; a low value is desirable for normal data and a high value for spoilage scenarios.

Table 6.1: Percentage of samples flagged in each scenario

Scenario	Isolation Forest (%)	Rules: high risk only (%)	Rules: moderate or high (%)
Normal (held-out)	3.0	5.0	37.0
Mid-layer hotspot	50.5	100.0	100.0
Humidity trapping	80.5	75.0	100.0
Gas build-up	51.0	71.0	78.5
Combined spoilage	100.0	98.5	100.0

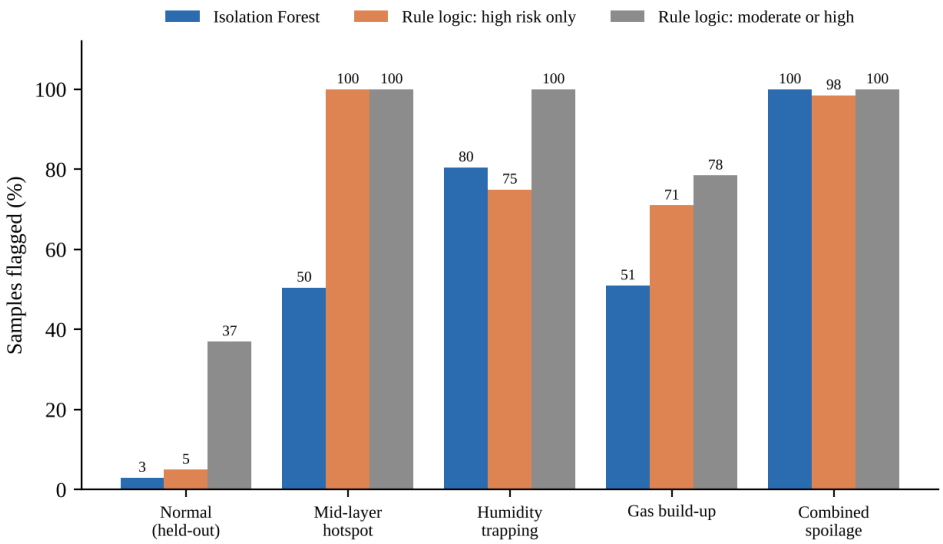


Fig. 6.1: Percentage of samples flagged by each detector in the simulated scenarios

6.3 DECISION-SCORE DISTRIBUTIONS

Fig. 6.2 shows the Isolation Forest decision scores. Normal samples lie at about 0.05 to 0.15 and combined spoilage samples between about −0.19 and −0.12, so they are clearly separated. Single-factor scenarios fall near zero (mean −0.007 for hot spot, −0.025 humidity, −0.007 gas) against +0.085 for normal samples.

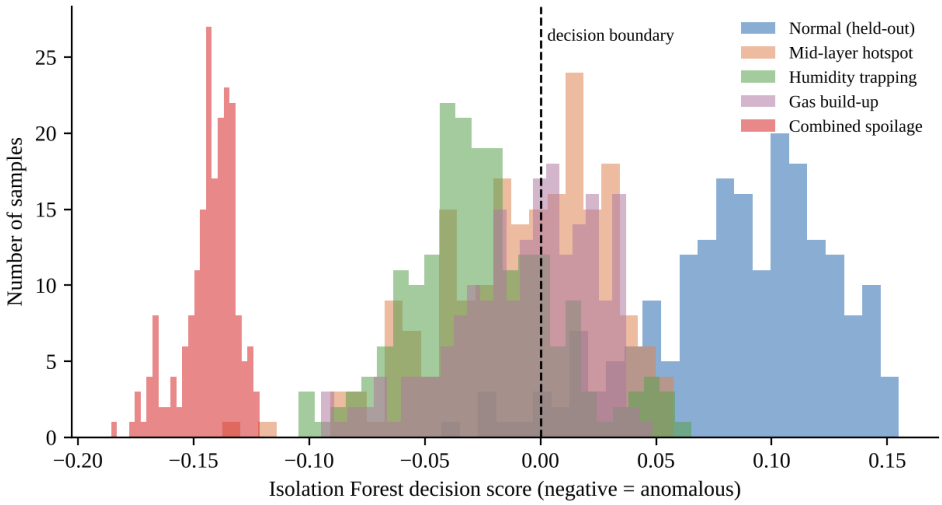


Fig. 6.2: Distribution of Isolation Forest decision scores for the five scenarios

6.4 DISCUSSION

Combined events are detected reliably: the Isolation Forest flagged all 200 combined samples and the high-risk rule 98.5%.

Single-factor events are harder for the model: it flagged about half of the hot-spot and gas samples and about 80% of humidity samples, since one feature moving away from normal changes the isolation score only moderately. The rules flagged all hot-spot samples as high risk, so combining rules (dependable for known critical conditions) with the learned model (better for joint drift) is attractive.

False alarms. On normal data the Isolation Forest flagged 3% and the high-risk rule 5%, but the moderate tier flagged 37% because ΔT above 3 °C is common in healthy synthetic grain; the threshold should be tuned on real data.

Cost. The prototype costs about Rs. 3,357 (Table 4.1), within reach of small operators, unlike the commercial systems of Section 3.4.

6.5 LIMITATIONS

- Training and test data are synthetic, and thresholds are not calibrated on a real trial.
- On-device scoring is threshold based; the Isolation Forest has not been exported to the ESP32.
- The firmware gas threshold (2000) is on the raw MQ-135 ADC scale and needs calibration, whereas the simulation uses a different scale.
- Grain moisture is logged but not used in the risk score, and only one BME280 is read.
- The public MQTT broker has no authentication, acceptable for a prototype only.

Chapter 7
CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

This project built a low-cost IoT node that senses temperature at three depths, humidity, gas and grain moisture, computes a three-level risk on an ESP32, alerts locally and streams to a smartphone over MQTT, at about Rs. 3,357.

In simulation the Isolation Forest flagged all combined spoilage events with 3% false alarms, while threshold rules responded more strongly to single critical conditions. Because deploying the model on a microcontroller proved difficult, a generated threshold-scoring header is used meanwhile.

7.2 FUTURE SCOPE

Extensions towards an industry-grade system include:

1. **Run the model on the device:** export the trees or use quantised TinyML.
2. **Rate-of-change features:** train on $\Delta T/\Delta t$.
3. **Incremental learning:** adapt the baseline to seasonal shifts.
4. **Better gas sensing:** a BME688, SGP40 or calibrated CO2 sensor [2, 9].
5. **Insect detection and self-diagnosis:** an acoustic sensor and sensor-fault codes.
6. **Autonomous power and range:** solar power with deep sleep, LoRa and push notifications.
7. **Automated retraining:** the pipeline of Section 5.5 with OTA updates.
8. **Aeration and field trial:** a relay-driven fan, a full-season trial and an authenticated MQTT broker.

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